

Three complementary and successive approaches to help collective decision-making: Budgetary Simulation, Business Game, Multi-Agent Simulation

Abstract

Many instruments of aid to strategic decisions are now operational and used by farmers and their counselors. The budgetary simulators are most frequent and techniques of optimization are mostly used in perspectives of research.

But these instruments meet limits. More and more problems: markets, utilization of limited natural resources cannot be considered on the farm level only but require a global approach.

The authors propose complementary approaches of increasing complexity and present experimentation results obtained by three types of instruments applied to the same type of problem.

They develop successively the utilization of a regional simulator aggregating the individual results, then the creation and the use of a business game. Finally then the possibilities of a multi-agent system will be considered.

Keywords: Group Decision Support System, Business Game, Multi-Agent Simulation.

1 Introduction

- 1.1 In the past, methods and instruments have been developed to help individual decision-making for farmers (Attonaty and Soler, 1992) (Attonaty *et al.*, 1999). Today, we are more and more confronted with problems where several actors are involved. These actors often have different interests. The aim is not to find an optimal solution as do models based on linear programming (Rieu *et al.*, 1994) or game theory (Thoyer *et al.*, 2001) but to create models which reach acceptable compromise by the different actors. This type of question is so frequent that it is not possible to answer personally all these demands. Our objective is to develop methods and instruments, usable at local level, to animate discussions between actors involved and to facilitate the emergence of acceptable compromise.
- 1.2 Problem finding is the first step. In general, a problem is defined by local actors such as public services confronted by a collective problem. We are not sure that this problem is the real one. The problem must be solved with full participation of the actors. A group of actors must be created to deal with the problem.
- 1.3 Problem solving is the second step. We think that both phases should be conducted together. Paradoxically, problem solving is a means of finding the real problem of creating a group of actors to contribute to the achievement of solutions. This group allows the real problem to emerge progressively using several phases: data collection, result analysis, building new scenarios to test. This phase of problem solving needs a validation step. This validation phase has two levels, the first one at elementary data level, and the second one at the level where instruments and models used should take into account the following features:
 - Ability to be understood by decision makers
 - Diversity of actor behaviour particularly “irrational behaviour”
 - Include evolution of actor behaviour due to their “learning by doing”
 - Show possible evolution of farms over several years including their possible disappearances
 - Emphasise economical, social and environmental consequences of decisions.
 - Include interaction between different actors.
- 1.4 In this way, we propose different approaches which are more and more complex, to use successively if necessary:
 - Individual budgetary simulation to help individual decision-making;

- Global budgetary simulation to help global decision-making;
- Business games to get new solutions with different actors;
- Multi-Agent Simulation based on previously collected data to imagine new scenarios and test quickly their consequences with different viewpoints.

2 An approach of Group Decision Support Systems

- 2.1 A Group of Decision Support Systems (GDSS) is broadly defined as an interactive computer-based information system which is used to enhance structured, semi- and unstructured group decision-making tasks in organisations. Like individual decision support system (IDSS), GDSS are designed to provide tools for decision-making by supporting the basic DSS functions of data, model and dialogue management (Sprague and Carlson, 1982).
- 2.2 This approach provides the (collective) group concerned with elements for reaching a compromise. On the one hand, this collective group plays an active part because it participates in the definition of the problem and in its reformulation. On the other hand, this group analyses results. At the very least, this group defines certain viewpoints to take into consideration. This approach is based on a budgetary simulator (Fig. 1).

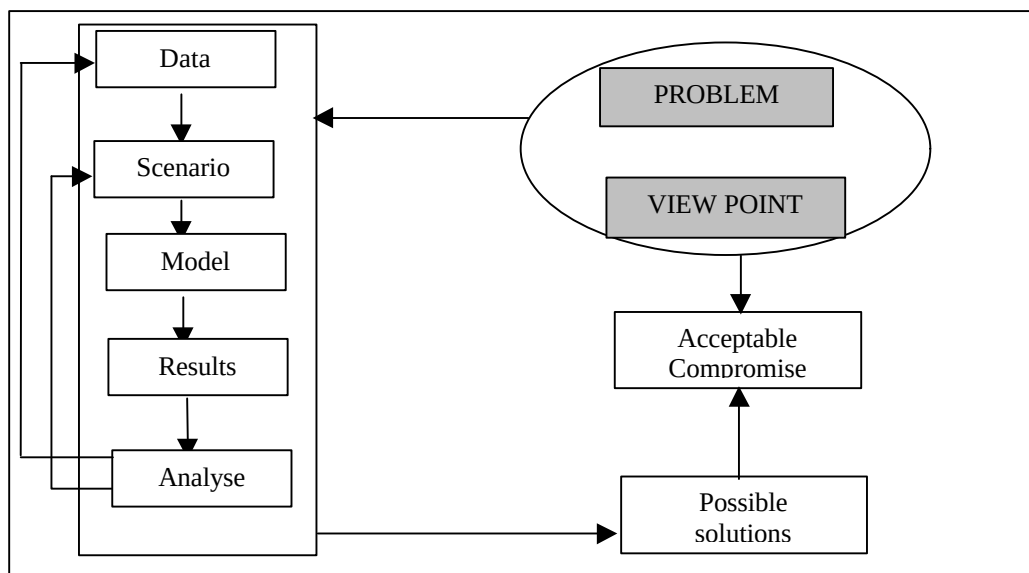


Fig. 1 : An iterative analysis of the problem.

- 2.3 Description of the model and its possibilities
 From a computer point of view, the Olympe simulator developed by J-M Attonaty (INRA Grignon, France) makes simulations for one or several actors, provides results and synthesises these results according to the needs of each actor (Fig. 2). On the one hand, this simulator allows the simulation of the three years before the beginning year of the simulation. Each actor can therefore compare the past given by the model with his own results. On the other hand, each actor analyses the results obtained by the model for n years (10 years max) with his own criteria (economics, work requirement, risk...).
- 2.4 Individual decision-making
 Two conditions are necessary for a satisfactory progress of the process: the participation of the actors involved and data quality. For this reason, before considering the global problem we have to pay special attention to certain individual actors in a classic phase of decision-making. When we have identified certain actors and their problem the process consists in collecting the necessary data and in an iterative way to modify the scenarios, quantify and analyse them (Fig.1, left side).

2.5 Advantages and disadvantages of this type of approach :

This first approach allows the actors concerned to develop criticisms and to propose improvements. this is the first way of model validation by users. Development of individual models makes it possible to build a first database at the local level. On the other hand, the farmers who are involved in this step are generally not representative because they are more dynamic and obtain better results, so the data representativeness is far from perfect. It is necessary to devote special attention to this point when we reach global level.

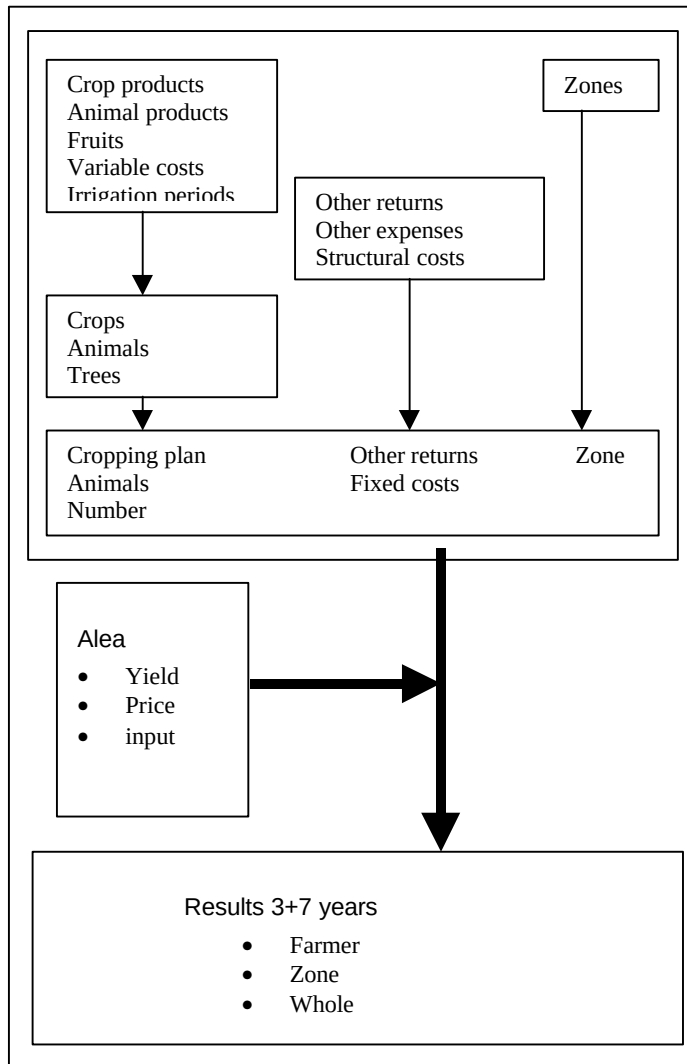


Fig. 2: The global model.

2.6 Passage to a collective level

The first step consists in building an actor typology taking into account criteria in accordance with the problem. For example: Zone, soil type, crop systems, farm size (Attonaty and Le Grusse, 1994). The second step consists in finding data, available at global level for several years, to compare results provided by the model with real data. Gaps between model results and reality are a way to detect key data to which special attention must be paid and complementary data collection must be made. The last step consists in defining pertinent criteria to analyse results. When the problem evolves criteria may also evolve. This process is iterative and interactive (Fig. 1).

2.7 An example of a regional simulation model: A simulator to help the negotiation in an irrigated perimeter in Tunisia.

2.7.1 The context

Tunisia is a country on the southern shore of the Mediterranean Sea which is especially concerned by problems of water management because it is reaching limits of its resources. This country has invested heavily in irrigation and today the main effort concerns management improvement of its irrigated area. We should note that, in an irrigated area, with the disappearance of subsidies and a very low price increase, the farmers' income decreases. The sugar beet crop was encouraged in the irrigated area studied. This crop is a big water consumer with no competitive production costs. In such a system, the interests of the different actors diverge over time. Dam administrators must manage the water resources, provide restrictions in order to avoid shortages and obtain satisfactory revenues. The sugar factory must assure its sugar beet provisions. Farmers want to have a better income and the Government in a more general manner prefers to valorise water resources. To build a negotiation framework for production choices in accordance with water availability, it is first necessary to imagine and evaluate different solutions and to search for negotiable solutions. To help the dialogue between the different actors involved (Figure3) we designed an area model to simulate different evolution scenarios.

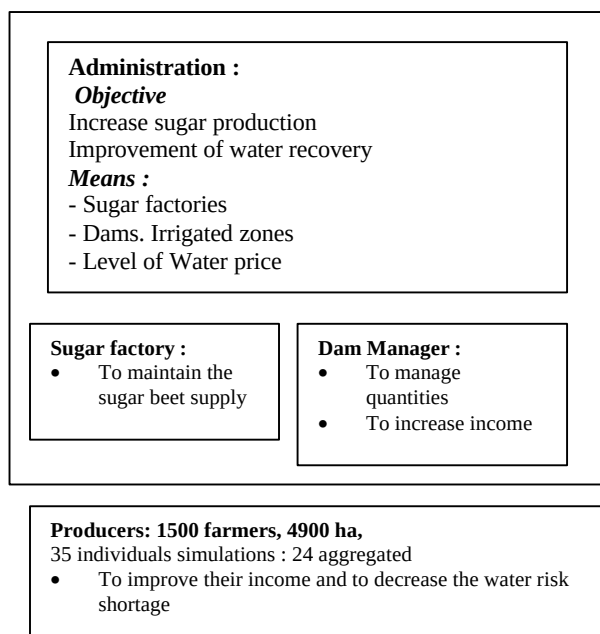


Fig. 3: A simulator to help the negotiation in an irrigated zone in Tunisia. Actors from the system.

Work (Ali, 1996) is carried out in three phases:

- Collection of technical data on farms as well as data gathering concerning their socio-economic environment;
- The design of a model of the irrigated zone that will be both a data base on systems and a simulation tool;
- Simulation of scenarios to increase dialogue between the different actors and to facilitate emergence of new solutions.

2.7.2 Typology and production Systems

The first phase consisted in establishing a typology of farms, by identifying one or several farmers from every type and building up with them different evolution scenarios of their farms. We used our budgetary simulator that runs over 9 years: 6 years of forecasting and 3 past years to validate the model. This phase allowed us to collect basic information on the different crops and to validate these

data while comparing past results provided by the model with the real results. This phase lead us to establish a typology of years and to define for every type of year inputs and outputs for every crop. The investigation was carried out on a surface measuring 662,7 ha, i.e, 14,5% of the sector surface and with 25 farms.

We determined three types of year in the beginning according to the rainfall of the year and water stocks in the dam during the agricultural year:

Year of type A: important stock in the dam and light rainfall.

Year of type B: weak stock and average rainfall.

Year of type C: important stock and heavy rainfall.

This phase based on individual interviews in farms also provided a lot of information on the constraints undergone by farmers and orientations that they wished to take.

2.7.3 Global model and its validation

Arranging a set of data on productions, the historical record of three agricultural years of the twenty-five farms investigated and with a lot of individual simulations we could develop a model for the sector. We regrouped all elementary data treated before reproducing the activity to the sector scale.

The computer program used comprises three parts:

- A technical part to define for farms from every zone (crops, animals, orchard), the inputs and outputs for different production levels.
- A part concerning the types of farms. Every type of farm has a name, is in a zone and has a weight. It corresponds to various farms seen earlier. We have first introduced cropping plan rotation, orchard, livestock composition, fixed costs achieved in the past.
- A part of risk definition. This results here from the definition of several types of year that entails variations on inputs and outputs. These risks are used in the calculation of last year's results to validate the model and in the establishment of the prospective scenario taking into account climatic variability.

This program provides, by choice, for a farm or for farms in a special zone, or a type, or for the set of farms of the whole sector, results in quantity and in value for the last three years and forecasting for six years according to the considered climatic scenarios (Fig. 3).

2.7.4 Model validation

For validation, we compared model results to available data on the zone. They come from zone statistics concerning crops surfaces and sugar factory productions for the sugar beet production and the dam administrator for water consumption. Concerning total surface, model results reach an average of 98% of the real surface with an inter-yearly variability that doesn't exceed 6%. But there are some big differences for some crops. The difference is very important for two crops. The shift is accentuated for the late potatoes and is explained by crop nature. It is a catch crop that is often forgotten by the farmer during the interview. The last remark that requires more reflection is the observation of sugar beet surfaces for the three agricultural years. This overevaluation oriented us toward the hypothesis that farmers overvalued sugar beet surfaces by distrust and to show that they respect regulation; Indeed the water price is a function of the sugar beet surface. A set of interviews verified this hypothesis and provided modifications entered into the model. The following figures (Fig.4, Fig.5) show real and calculated surfaces and productions before and after validation. For water consumption, the model provides in the first place a somewhat lower consumption than the reality. A detailed analysis completed by interviews showed that in years of high rainfall where the stock in the dam was important, the farmers irrigated less than the crops needed. For the dry years but with an important stock in the dam there was a very heavy use of water. The model results coincided with a normal rainfall with weak reserve. In this case, the irrigation rules were respected by restriction measures imposed by the administration. In a general way, we could note a tendency to water wastage in the absence of shortage of resources. This is due to the farmers' practices and to uses by non-farmers and also to losses in the network. It has been restored in the model like shown in fig.6:

Crops	Statistics data	Before Validation	%	After validation	%
Wheat	2050	1890	92	2004	98
Sugar beet	984	1162	118	1063	108
Early potato	207	133	64	159	77
Late potato	214	128	60	128	60
Total zone	4695	4609	98	4608	98

Fig.4: Surface comparison between model and reality before and after validation (in ha)

Crops	Statistics data	Before validation	%	After validation	%
Wheat	8477	6567	77	7067	83
Sugar beet	48204	58200	121	54220	112
Early potato	3122	2183	70	2581	77
Late potato	2635	1350	51	1350	51

Fig.5: Production comparison between model and reality before and after validation (in ha)

Designation	A	B	C	Average
Results of the model	10,5	7,8	5,1	7,8
Other uses	0,5	0,5	0,5	0,5
Losses (15%)	1,65	1,25	0,84	1,25
Total re-established	12,65	9,55	6,44	9,55
Invoiced real consumption	14,02	9,71	5,83	9,85
%	90	98	110	97

Fig.6: Water consumption in the sector for each year (in Mm³)

2.7.5 Simulations to enter into a dialogue between actors

The different years studied constitute different types that we qualified by rainfall amount but also by water available in the dam. To be able to affect frequencies to these years we analysed rainfall recorded over 33 years. We could bring to light the following results: year - type A (25%), year - type B (50%) and year - type C (25%). In the zone we identify three main actors whose interests are different. The dam management administration who must balance their accounts and is interested in water consumption and water pricing. The sugar factory which is interested in the quantity of sugar beet produced, and farmers of the sector interested in their revenue. Furthermore, extension services provided data collected at the farm level corresponding to more intensive production. To help negotiation between these three actors we established different evolution scenarios while considering different cropping plans and different intensification levels.

Scenario 1: The possibility of increasing sugar beet production.

The sugar factory has an important role in this region and tries to increase the sugar beet yield. Different more intensive techniques are known but they require more water and this presents problems.

We simulated different scenarios with an increasing intensification. The synthetic results presented here (Fig.7) show the advantage for the sugar factory and for the water supplier who increase their revenues. However the water supplier risks not being able to provide the amount of water needed

every year. On the other hand, farmers obliged to intensify their crops do not benefit from their efforts: their revenue increases very little.

Scenario	Water need	Water fees	Sugar Beet	Farm income
Actual	100	100	100	100
Intensification II	115	115	114	102.5
Intensification III	128	128	125	104

Fig.7: Consequences of sugar beet intensification. Average results. (basis 100 to the present situation)

Scenario 2: Sugar beet suppression

Following these first simulations farmers and the dam manager are wondering about the value of maintaining sugar beet. To this end, we established with local agronomists different crop rotations including pulses and vegetable-field crops. Results (Fig.8) show a strong decrease in water consumption. Dam revenues decrease a little because the water price is higher: it doesn't benefit from the reduced price allowed for sugar beet crops. Therefore, the dam administrator is interested by this solution because his income is slightly lower and he can consider selling water to other irrigated zones. Farmers are also interested: they discover that no cropping sugar beet does not decrease their income very much. The conclusions of these simulations demonstrates the value of intensification.

Scenario	Water need	Water fees	Sugar beet	Farm incomes
Actual	100	100	100	100
Variant 1	64	83	0	98.6
Variant 2	64	83	0	97.4
Variant 3	64	83	0	96
Variant 4	64	83	0	94.8

Fig 8: Consequences of sugar beet suppression. Average results for actual level of intensification. (basis 100 to the present situation)

Scenario 3: Consequences of different levels of intensification

We simulated two intensification levels for different cropping plans. Results (Fig.9) show the interest for the dam manager and for farmers in intensifying or suppressing sugar beet. To survive, the sugar factory should increase sugar beet prices but it cannot do so without increasing its production cost or receiving an important grant-in-aid. The sugar cost increase would entail a loss of its outlets.

Scenario	Water need	Redevance	Sugar beet	Farm incomes
Actual with sugar beet	100	100	100	100
Level I	64	83	0	98.6
Level II	84	108	0	109.3
Level III	104	133	0	116

Fig.9: Intensification consequences for the variant 1(basis 100 to the present situation)

Consequences of these three scenarios are synthesised in the figure 10.

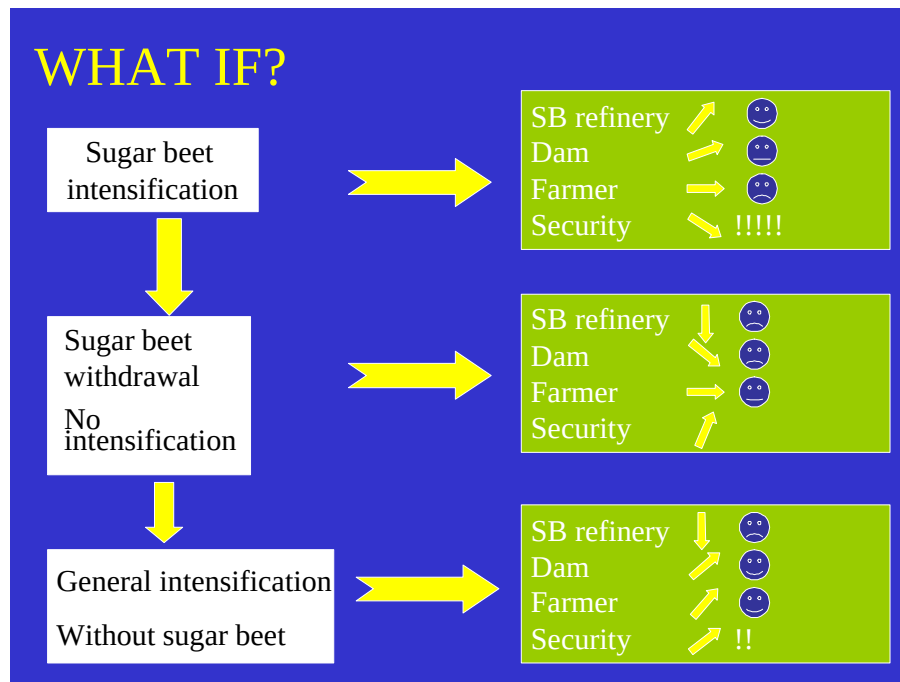


Fig. 10 : Consequences of scenario tested

Scenario 4: Results sensitivity of price variations.

The previous results are directly produced by the model. But they are based on price hypotheses and are therefore questionable. So we used another simulator function to do a sensitivity analysis: we progressively decrease the gross product from other crops until the solution results with sugar beet are reached.

This analysis showed that for the different intensive variants the sensitivity is weak: it is necessary to lower gross product from 31% and 45% according to intensification level to obtain the same income as solutions with sugar beet.

2.7.6 Possible extension of this approach

For an outside observer, the results seem clear; sugar beet must disappear. In fact, the decision is incumbent upon the local actors: the model only provided negotiation elements. It is based on a farm typology and a detailed analysis. The design of a global model of the zone and its validation by comparison to the existing global data showed the limits of individual interviews and made it necessary to carry out complementary interviews concerning the actual cropping surfaces and irrigation practices. Once this sufficiently validated model could be interrogated, it informed the different actors progressively of the consequences of crop rotation modifications and intensification levels.

The presentation of this model to the actors revealed the necessity of primarily presenting in detail the data used in the model. It is only after having shown and discussed inputs and outputs of the main crops and presented the typology used that global results can be presented and discussed. The conclusion concerning the low profitability of sugar beet has been shared by participants who now, some years after this study, no longer have a sugar factory. The working group created on this occasion still exists and has requested a new simulation to refine other decisions. Other irrigated zones need similar studies and we try to respond while training local students to understand local problems and local agricultural practices. This work results in fact from an experiment carried out by the authors and a Tunisian student and required six months of work. It seems therefore easily repeatable and several comparable studies are currently in progress.

The computer program used was a prototype. Its use showed its strengths and its weaknesses. A more elaborate version is now available.

Teaching program and methods is now an important part of the course of lectures given in IAMM (Montpellier, France). Many studies of this type have been successfully carried out by students from different countries in their own country after completing this course.

3 An approach based on business game

3.1 Previous experience showed the necessity of developing methods and instruments in order to construct scenarios quickly with the decision-makers so that their questions could be answered and to modify these scenarios in accordance with the deciders' thoughts. This also showed us the opposing interests of different actors and existence of different coalitions.

3.2 Following comparable studies conducted with the same approach and the same instrument, we tried to use the data collected and the same simulator to create a business game. This tool puts players in a situation where they must make decisions forcing them to negotiate in order to establish a production plan taking into consideration global constraints and the other players' decisions. The game objective is that actors can learn in the " virtual " situation to act better in the "real one". The first game users were students and the case study is a simplified case inspired by a Mediterranean zone. In their cursus the students had already constructed a linear programming model of this zone and this game was also the opportunity to recall the underlying strong hypotheses of this technique.

3.3 The context of the game: Different production systems in an agricultural region

The virtual region includes four main systems with different potentialities, technical levels and farm numbers.

- The system A has a lot vegetable-field crops with 500 farms of 10 Ha each;
- The system B has vegetable-field crops and cereals with 100 farms of 50 Ha each;
- The system C has vegetable-field crops and orchards with 100 farms of 30 Ha each;
- The system D has cereals and fruits with 50 farms of 200 ha each (orchard: peach trees 5 ha and apple trees 25 ha).

Every production system has production factors, crop potentialities and production techniques according to three intensification levels. The main inputs are: land, workforce, and water quotas with different prices. The agricultural region is also determined by production history, by the evolution of the consumption and the availability of production factors, as well as the evolution of markets and prices evolution. Using the game, the water availability is uncertain and is randomly fixed every year. The different products have a function that joins price and global quantity produced with different elasticity according to products. All these elements are unknown by the players who discover them progressively.

3.4 Progress of the game

Every type of agricultural farm is "managed" by a student group. This group knows all the technical potentialities of its production system and those of other farms as well as the historical record on a regional level.

Then every group must decide and must define its production plan without knowing the decisions of the other groups in the past and what they are going to decide for next year. No group knows either the precise evolution of availability of the production factors outside the farm, production quantities and the markets prices.

The game is going to take place step-by-step:

Every group decides a cropping plan according to its technical and financial capacities, its water and workforce availability. All decisions are aggregated in the regional simulator and the total production is calculated in accordance with the climatic conditions randomly selected. Market models defined by product calculate quantities sold and selling prices on different markets. In return these elements enable the global income to be calculated for every farm. Then the results are given to every group

which must then take a decision for next year. These results contain group farm results and regional results. The game is structured in two phases:

- A first phase without communication between players;
- A second phase where players can discuss and negotiate

First phase: no communications between players (Fig 11)

The first game period is characterised by over production on certain products and deficits on others. Water and manpower availability are immediately reached, and deficits appear; this is due in general to the use of intensive production levels which theoretically brings more cash. Individual optimisation models based on linear programming used by groups reinforce the search for highest results and therefore increase the intensification process. Excess product prices fall and those of deficit productions increase strongly. In the following periods we saw the emergence of production diversification to minimise risk. Water consumption decreased and was more or less in accordance with availability. Farms and regional results were improved, but surpluses and important deficits persisted.

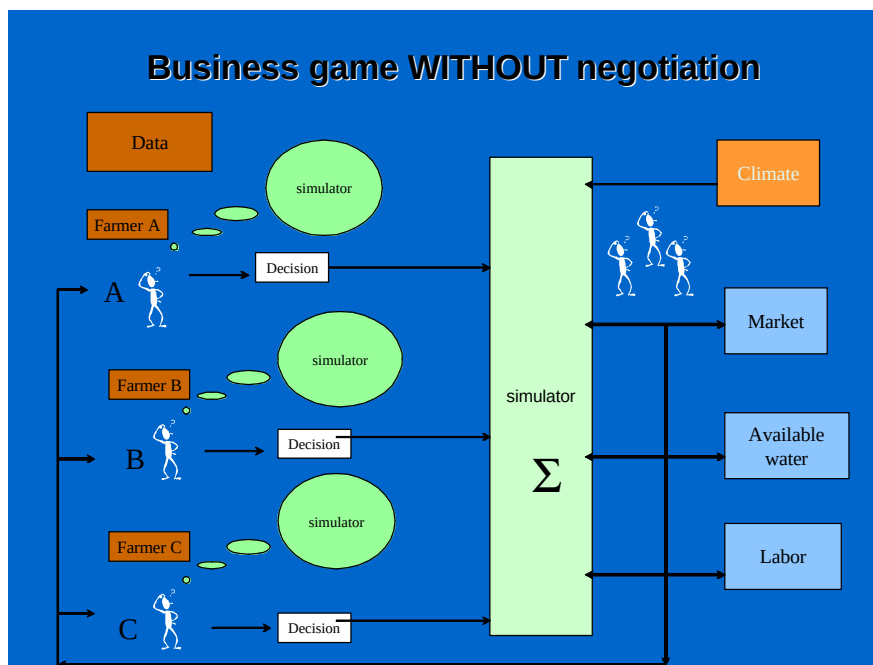


Fig. 11 : Business game without negotiation

Second phase: possibility to discuss and negotiate (Fig 12)

This second phase started by a first meeting between players and information was exchanged about the different production systems. A common analysis of the region and the evolution of markets and resources was made. This first exchange enabled the decrease in information asymmetry between actors, but failed on two points: determination of the items which it is possible to discuss and how to discuss. In the following periods we noticed the progressive emergence of agreements between players: definition of quotas for some products, negotiated monopoly creation for a few products. Lastly, a water market appeared. These negotiations brought a strong reduction of surpluses and an increase of farm revenues.

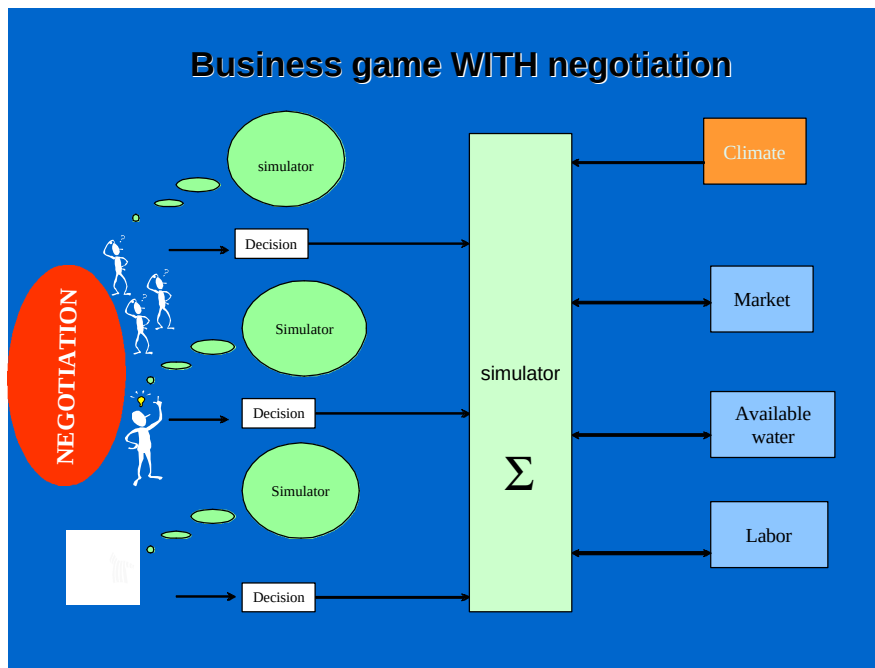


Fig. 12 : Business game with negotiation

3.5 Learning from the results of the business game

The game experiment taught several lessons:

Without negotiation the individual training allows the improvement of global performances, but a maximum level quickly appears in a system where resources and markets are limited and uncertain.

Use of a negotiation process enables a strong improvement of system performances up to a new level.

In parallel, an optimisation model of the region was used and provided for each period the optimum level that the region could reach if one had a perfect knowledge of available resources and markets.

The comparison between the optimal solution (regional linear programming model) and the results of the simulations show there is still an important progress margin after the second phase. We think that the game continuation would have allowed improvement. But due to lack of time we were not able to verify this hypothesis. Additionally, we think that an elementary training in game theory would have allowed players to progress more quickly. Also, this hypothesis will have to be tested.

3.6 Conclusion

In a context where markets and prices change rapidly and production techniques are moving, farmers are forced to react quickly. A lot of more or less sophisticated methods and instruments are now at the disposal of extension services and used to a variable extent. But help for individual decision-making is limited as many problems occur rather at a wider level than at an individual one. So, markets, environmental conservation or natural resource management must be considered at a more global level and must lead to a dialogue between different groups of actors with contradictory interests. In this perspective simple techniques as the simulation used in an interactive and progressive way with decision-makers can bring interesting results at low cost. One of its advantages is that it is easy to understand: the computer only plays the role of an improved calculator and data used are easily visible. The different experiments that we conducted showed us that they could be used, after a short training, by farm advisers or students. In this case its use on concrete problems by decision-makers proved to be an excellent training to a global approach.

The business game is based on the same type of data and on the use of the same computer program. It constitutes a natural extension of the previous approach to show the co-ordination necessity between different decision-makers and the value of negotiating to establish a set of applicable rules. If it is very formative but it seems rather time consuming. Its use by students is possible but the possibility of its use by with real decision-makers needs to be demonstrated.

In fact, these different approaches are very complementary and can be used successively. The initial data collection phase and their validation are a fundamental step. Once in possession of such data it is possible to create a decision-makers working group asking questions and analysing answers. So a real dialog can start on the real problems and avoid solving false problems in an academic way.

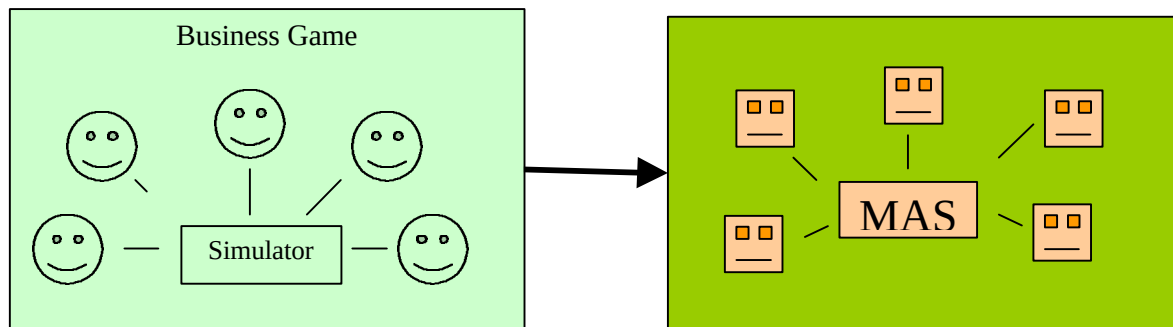


Fig. 13 : From the business game approach to the approach based on multi-agent

4 An approach based on Multi-Agent Simulation

- 4.1 The business game presented above provides interesting results but it requires a great availability of the actors because it requires several days. Actors learn rules and behaviours fast, so it is hard to replay the game many times.
- 4.2 An approach based on multi-agent would make it possible, by simplifying the actor behaviour defined in the business game, to make several simulations.
- 4.3 The objective of the model is to show the importance of regulations by simulating over a number of years the evolution of a group of farmers who use a limited water resource.
- 4.4 In the example presented below, we have considered a virtual zone where a problem of sharing water takes place (water stock considered as a water table). Each farmer agent will negotiate a water amount with the water supplier agent. Each farmer agent has a different behaviour towards water request.
This resource is managed by a water supplier who allocates water to each farmer in accordance to the regulations. The amount of water consumed depends on the climate of the year, the irrigated area and the level of irrigation. The crop yields are related to rainfall and water allocation.
- 4.5 We considered that the actors of the water allocation process are the following:
 - The farmers: each of them has cash money, order of priority on crop preferences, objectives and an attitude towards risk. He has a crop area divided into plots and his own irrigation capacity. Each year he decides on his cropping plan, which must respect his crop preferences and agronomic rules. This order of priority is partially related to crop yield, which depends on the climate of the year (unknown when he takes his decision) and on the water available. Possibly at this time he could negotiate with the water supplier for the water he needs. At the end of the year, after his economic results, each farmer can modify his crop preferences and decide whether or not to invest in more irrigation capacities.
 - The water supplier: each year he has an amount of water that he allocates to the different farmers. He has expenses and must balance his accounts.
 - The information supplier: each year he receives the individual farmers' economic results, analyses and classifies them and sends a global report to each farmer.
- 4.6 We do not intend to provide the optimal solution but only to show real negotiators the possible consequences of the water allocation rules they have decided on, according to different criteria:

economic (global output), ethical (disparities between agents), environmental (water savings). (Figure 14)

- 4.7 We intend to use this model in the future as a way to improve negotiation to determine water regulation at a local level.

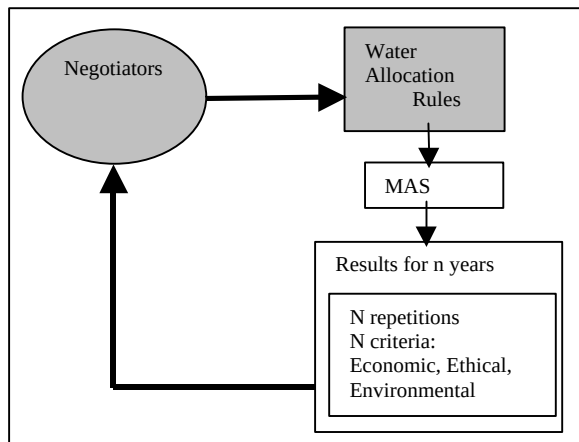


Figure 14. Model role in the negotiation

4.8 An overview of agent notions

Many definitions of an agent are based on the notion of strong AI-based agents. Even if there is no real agreement as to what the term agent means, some key features have been identified (Ferber, 1995) (Doran, 2001) (O'Hare and Jennings 1996)(Weiss 1999). Wooldridge (Wooldridge, 1999) proposed a definition of an agent as "a computer system, situated in some environment, that is capable of flexible autonomous actions in order to meet its design objectives". As pointed out in (Wooldridge and Jennings, 1995), autonomous agents and multi-agent systems represent a new way of analysing, designing and implementing complex software systems.

4.9 Advantages of the multi-agent approach

As (Brandolese *et al.*, 2000) and (Doran, 2001) have pointed out this type of approach offers a series of advantages, in that:

- They constitute a natural way to handle logically and physically distributed problems.
- They need neither global control nor global data storage.
- They enhance the overall reliability (capability to recover from the failure of individual components, with graceful degradation in performance) and robustness (tolerance to uncertainties in data and knowledge).
- The modularity of such systems allows their extensibility.
- Their architecture is simple and the way they work is transparent.
- They take into account many agents with different behaviours.
- They enable agents to learn and to change their behaviour in the light of new information.
- They simulate agent evolution over a long period in terms of growth and bankruptcy.
- They consider alliances and lobbies linked to exchanges between agents.

4.10 Exemple of application

The simulator MANGA (Multi-Agent simulator to help NeGotation in the wATER management) allow to analyse the evolution of a set composed of agents who use a limited water resource (Le Bars and Attonaty J-M., 2001).

4.11 The agents in the model

Our system does not have a fixed central control. This means that there is no specialised control agent and each agent is able to try to obtain an amount of water via negotiations taking into account individual preferences, agent rationality, and different objectives. Our modelling approach takes into account two types of agents (Fig. 15): Cognitive Agents and Reactive Agents.

Agents	Number	Type
Farmers	n	Cognitive
Water Supplier	1	Cognitive
Information Supplier	1	Reactive
Crops	n	Reactive
Climate	1	Reactive

Fig. 15: Agents in the model developed

4.12 Cognitive agents

In our model, we define farmers agents and the water supplier agent as cognitive agents. If we take into account farmer agents for example, their beliefs are represented by knowledge divided into several parts:

- A database containing farm area, list of plots and the crops cultivated in previous years, crops classified in order of preference based upon profitability and easiness, cash amount, etc.
- A database containing calculation procedures for water requirement, crop yields related to rainfall and water allocation, income and cash, etc.

Their desires are represented in a database containing decisional procedures made up of a set of production rules to determine cropping plan, irrigated crops and water amount, size of irrigated area and corresponding investment, crop preference order, etc.

Their intentions correspond to a set of rules to enable each farmer to reach its own goal.

At this stage of the model we have modelised different farmer behaviours in relation with their water need, the type of year reference (Wet, Dry or very Dry) for each farmer agent and a percentage of their ability to bluff. Two farmer agent behaviours have been considered: (i) the first one (A) is greedy and selfish: the farmer agent asks for 120% of its water needs in very dry years; (ii) the other one (B) is reasonable: it only asks for its water needs in dry years.

4.13 Reactive agents

In the model, the structure of reactive agents (crops, climate) is only made up of knowledge and communication modules. Each farmer agent has a set of knowledge given as inputs for each simulation. This set includes (i) a database containing data: farm area divided into plots, crops classified in order of preference based upon profitability and water consumption, (ii) a database containing calculation procedures for water requirements, crop yields related to rainfall and water allocation, for example and (iii) a database containing decision-making procedures made up of a set of production rules to determine cropping plan, irrigated crops, water amount, among others. Since these agents have different objectives, such as increasing or not increasing their revenue and their security level, their strategy module contains rules which can evolve over time with respect to new information and better knowledge of other agent behaviours. The agents decide how to negotiate, what crop yield objective should be reached according to climate conditions, and the cash to invest. Lastly, our agents can communicate in accordance with the type of information exchanged. If the information is private each agent can send a message to the mailbox of a receiver, which has the means to process it. In the same way, it can receive a message in its personal mailbox and process it. Otherwise, the information is public or semi-public. This type of information can be processed by agents which have the method necessary to retrieve it.

4.14 Simulation

The simulation is carried out over a number of years (12), each of which is made up of four sequences.

Determination of the cropping plan

- Each farmer agent determines its cropping plan and its water needs. This is an iterative procedure: a farmer agent determines a first cropping plan, fixes its yield objectives for a climatic type of year according to its behaviour and calculates its global water needs. This request is made at a time when neither the climate of the year nor the other water requests are known.
- The water supplier agent receives a set of water requests, adds them up and proposes an amount of water to each farmer agent in accordance with the water allocation rule being tested.
- Each farmer agent receives information about the amount allocated to it. These exchanges stop when the water supplier agent has no more water to allocate or no new requests. The water supplier agent determines its answer in accordance with the regulation tested.

Climate determination and crop growth

The climate of the year is determined at random. In accordance to the type of climate each crop agent has a method to select the water answer curve to use to calculate yields.

Economic results

- Each farmer agent calculates its yields and its economic results. It sends its results to the information supplier, which synthesises the information coming in from each farmer agent.
- The information supplier agent provides each farmer agent with global information concerning the highest, average and lowest revenues and crop yields.

Decision-making

Each farmer agent decides whether or not to invest in more irrigation capacities and possibly to change its behaviour.

4.15 Results

In MANGA, we analyse results from different viewpoints:

- An individual viewpoint, where we can observe water needs and water allocated per hectare in accordance with their behaviour.
- A global viewpoint, where we consider global corn production.
- An ethical viewpoint, where we focus on yield disparities between the different groups of agents with their different behaviours.
- An environmental viewpoint, where the amount of water allocated, used and left unused is represented.

4.16 Rules tested

Scenario1: Rule based on water requests pro-rata

“If the water amount requested is less than the water available,

Then each farmer agent can use the water amount requested.

Otherwise water is allocated as a pro-rata of the amount of water requested.”

Figure 16 shows the results we obtained for an individual viewpoint (water allocated per agent) with varying individual behaviour. Taking into account the water amount requested by farmer agents and the behaviour of each farmer agent (A or B as described in **section 5.5**) we have fixed a water amount that is a little higher than the water needed in a dry year. As we expected, when all the agents have the same modest behaviour, not all the water available is allocated (Figure 16 (b)). We also see that when we have a farmer agent population made up of farmer agents with different behaviours (figure 16(c)) and when we take into account the rule based on water requests pro rata, the water is allocated to

farmer agents who do not use it (Behaviour A) and this penalises farmer agents with modest behaviour B. Figure 18 gives the results using the same rule but taking into account the global viewpoint, i.e. global corn yield for all farmer agents. It should be noted that as was the case in figures 16(a) and 16(b), since farmer agents have enough water for wet and dry years they obtain the maximum corn yield for this type of year. We can also observe that they do not obtain the maximum corn yield in very dry years because of lack of water. It is interesting to note that in the absence of reasonable agents (behaviour B), global yield is higher but heterogeneity of requests leads to greater yields as shown in figure 17. Furthermore this heterogeneity introduces a global reduction in yield, and a higher yield disparity (Figure 18).

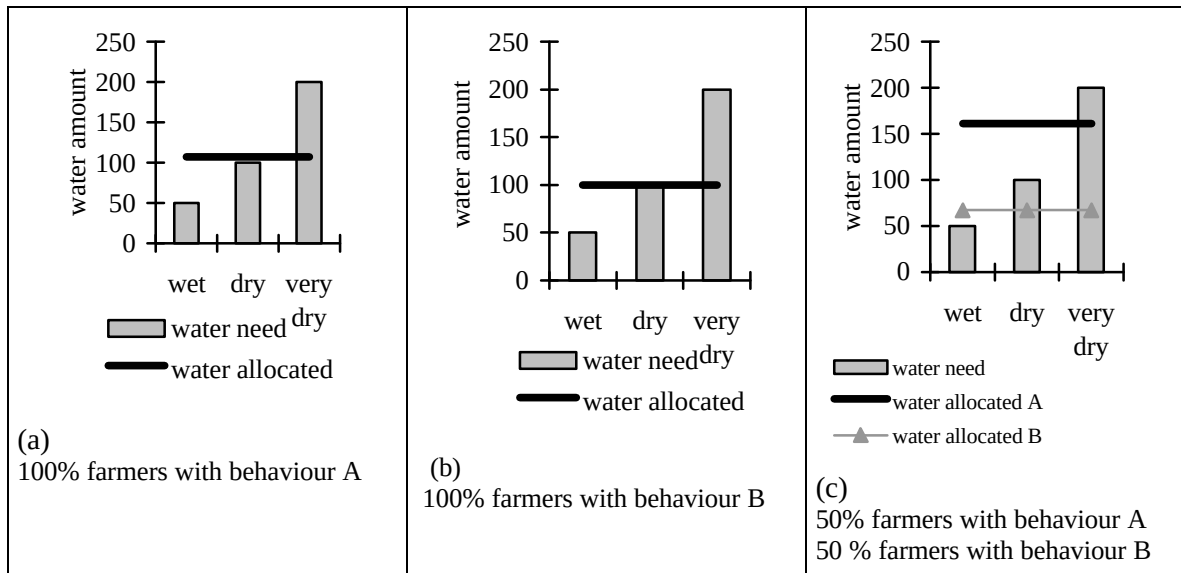


Figure 16: Results for individual viewpoint with rule based on water requests pro rata

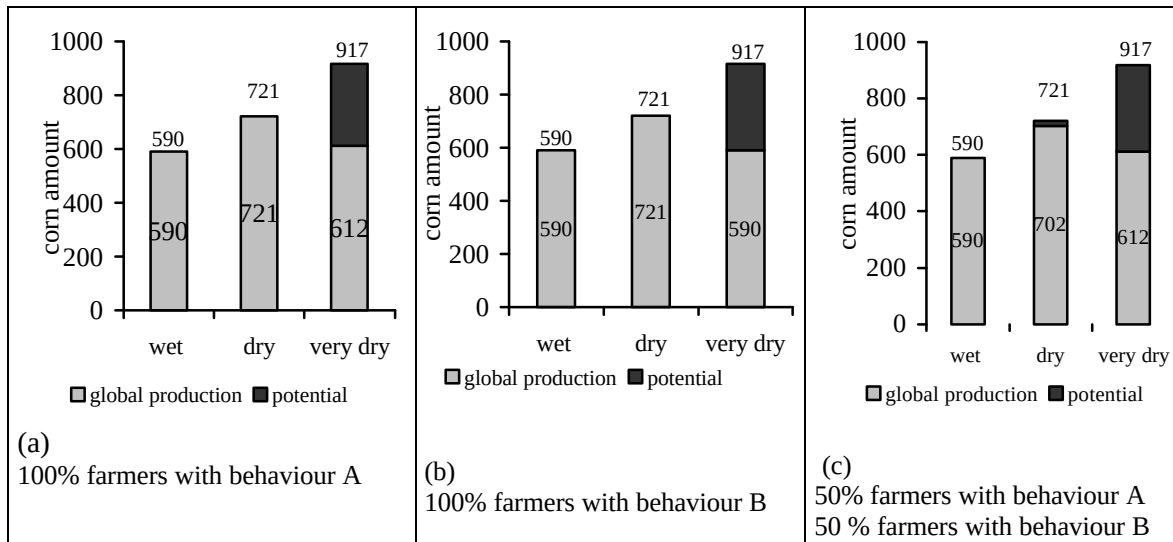


Figure 17: Results for global viewpoint with rule based on water requests pro rata

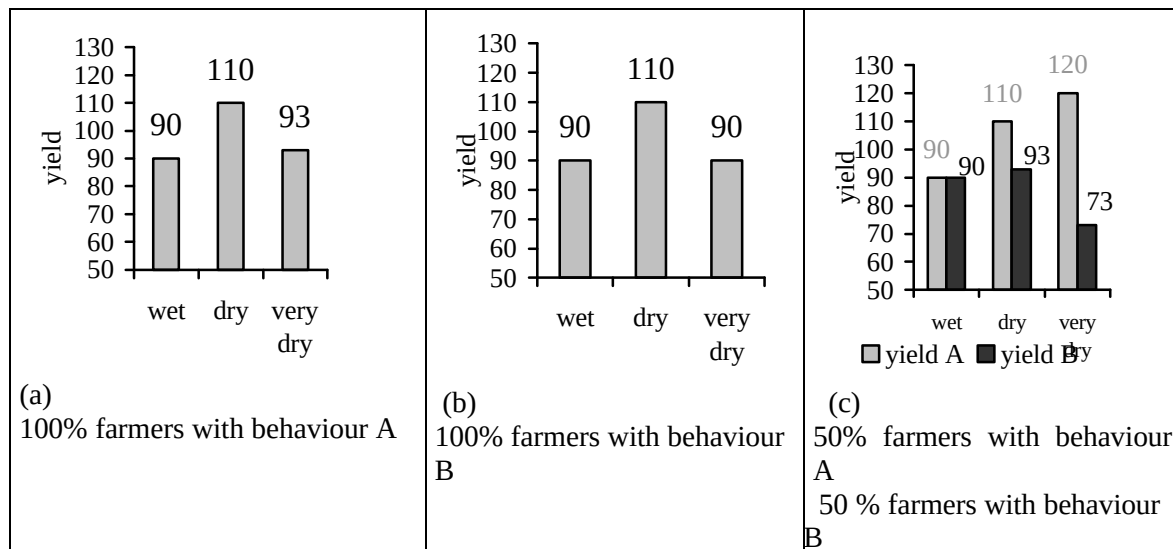


Figure 18: Results for ethical viewpoint with rule based on water requests pro rata

Scenario 2: Rule based on crop areas irrigated

*“If the water amount available is more than the water amount requested,
Then give each farmer agent the water requested,
Otherwise water is allocated in function of crop areas irrigated.”*

With this rule we obtained better results (Figure 19) in the short run in terms of global production and disparities. The results are similar to those obtained with a rule based on a centralised economy. With regard to this situation, which translates field observations, we tested a more authoritative rule that allocates water according to irrigated areas. In this case, we obtain maximum yield and absence of yield disparity.

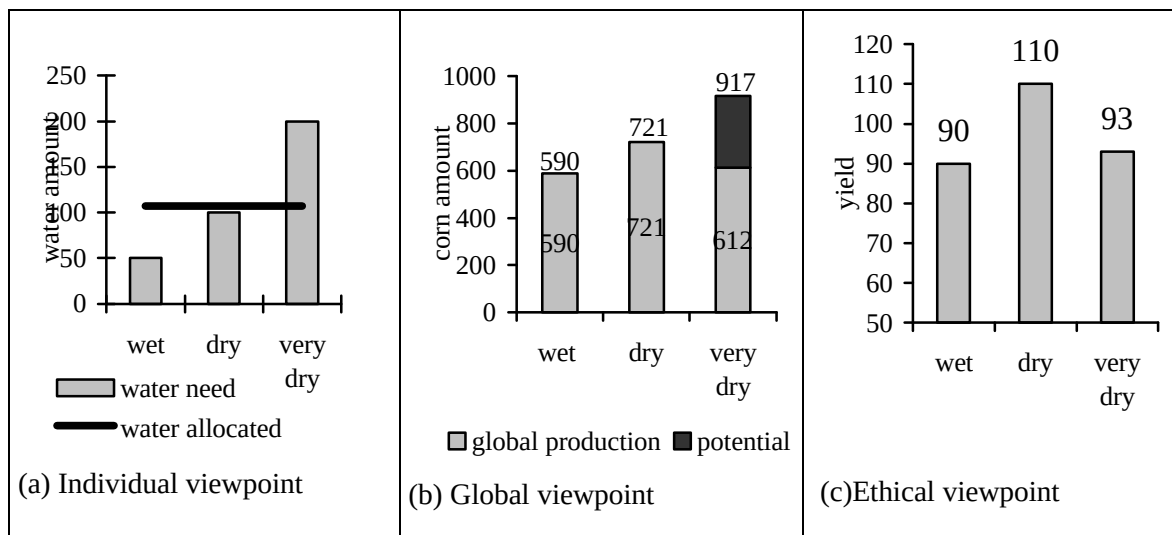


Figure 19: Results for global viewpoint with rule based on crop areas irrigated

Scenario 3: Rule based on water requests pro rata and knowledge of water allocated to the others

The farmer agent has an adaptive rule; it knows the water allocation of each farmer and asks for the maximum amount of water allocated. Results (figure 20) show that we get the same results in this bottom-up evolution, based on better knowledge as in the top-down way of fixing water allocation (rule based on water requests pro rata).

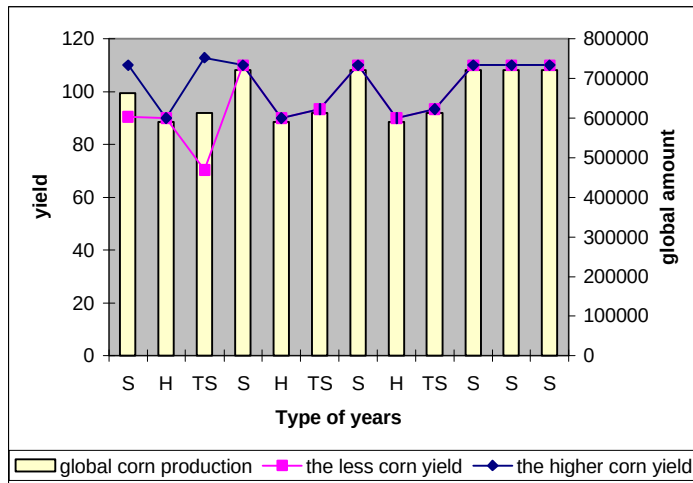


Fig 20: Results for rule based on water requests pro rata and knowledge of water allocated to the others

5 Conclusions

- 5.1 Over the past few years, many instruments have been developed for individual decision-making. Optimisation and simulation models have been used frequently. The choice between these approaches depends on the one hand on the user-friendliness of the tools and on the other hand on basic assumption made about decider rationality.
- 5.2 In a context where markets and prices change rapidly and production techniques are evolving, farmers are forced to react quickly. A lot of more or less sophisticated methods and instruments are now at the disposal of extension services and used to a variable extent.
- 5.3 But help for individual decision-making is limited as many problems occur rather at a wider level than at an individual one. So, markets, environmental conservation or natural resource management must be considered at a more global level and must lead to a dialogue between different groups of actors with contradictory interests.
- 5.4 In this perspective simple techniques such as the simulation used in an interactive and progressive way with decision-makers can bring interesting results at low cost. One of its advantages is that it is easy to understand: the computer only plays the role of an improved calculator and data used are easily visible. The different experiments that we conducted showed us that they could be used, after a short training, by farm advisers or students. In this case its use on concrete problems with decision-makers proved to be an excellent training to a global approach.
- 5.5 The business game is based on the same type of data and on the use of the same computer program. It constitutes a natural extension of the the previous approach to show the necessity for co-ordination between the different decision-makers and the value of negotiating in order to establish a set of applicable rules. If it is very formative but it seems rather time consuming. Its use by students is possible but the possibility of its use with real decision-makers needs to be demonstrated.
- 5.6 The time constraint seems otherwise less restricting with Multi-Agent models. It is possible to establish several result sets taking account of the environmental variability and to test a limited set of rules for a ressource assignment. A first test showed the possibility of such an approach that is still in the research field.
- 5.7 Consequently, we have shown in this paper that these tools: budgetary simulation, business game and multi-agent approaches are not in opposition but must be used in order and according to the problem. The problem is not clearly defined in the beginning but evolves progressively: the use of tools can help the real problem to emerge progressively.
- 5.8 Different tools are now available: some are already operational and are in use, others are in the research field and must be tested and improved.
- 5.9 But whatever the quality of the tools to solve problems, their use is meaningless if we do not benefit from the support of a group of actors concerned by the problem at different levels and by the evolution formulation of the problem.

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